

Supplementary Material for “A Homogeneous Compensation Branch for Indirect Evolutionary Rescue in a Reduced Free-Space Predator–Prey Model”

Draft author list to be finalized

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S1. Numerical classification protocols

The main manuscript reports consolidated outputs from the numbered analysis scripts rather than new simulations. Classification used the same vocabulary across ODE and PDE outputs:

- **Persistent:** predator density remains positive and late-window change/residual diagnostics indicate a steady or effectively settled state.
- **Extinct:** predator density approaches the extinction threshold and late-window diagnostics do not indicate recovery.
- **Transient:** the trajectory does not meet persistent or extinct steady criteria within the tested horizon, often because late-window change or residual diagnostics remain non-negligible.

The tail-window logic was used to avoid treating long transients as equilibria. For PDE tests, classification of spatial means was paired with spatial coefficient-of-variation diagnostics for n , w , and q . A basin label was compared with a matched homogeneous control when heterogeneous perturbation effects were evaluated. The classification logic is diagnostic, not a proof of global basin structure.

For the no-evolution counterfactual and evolutionary-rate sensitivity diagnostics, the supplement distinguishes two persistence notions. *Deterministic tail-based persistence* is the baseline mathematical classification from the terminal tail window. *Threshold-safe persistence* additionally requires that $w(t)$ never cross below the extinction threshold during the tested horizon. This threshold is a deterministic diagnostic/quasi-extinction convention, not an empirical extinction probability. The distinction matters because a deterministic ODE trajectory can recover after a very small predator density, whereas an ecological population may be effectively lost after crossing an extinction threshold.

S2. Mathematical appendix: ODE Jacobian and Routh–Hurwitz margins

Write

$$\begin{aligned} F_n &= nR, & R &= r(q)z - \xi - a(q)w, \\ F_w &= wP, & P &= b(q)nz - (m + s) - \mu w, \end{aligned}$$

$$F_q = \nu q(1 - q)G, \quad G = \Delta r z - \Delta a w,$$

with $z = \kappa^{-1} - n - w$, $\Delta r = r_v - r_u$, $\Delta a = a_v - a_u$, and $\Delta b = b_v - b_u$. Since $z_n = z_w = -1$,

$$\begin{aligned} R_n &= -r(q), & R_w &= -r(q) - a(q), & R_q &= G, \\ P_n &= b(q)(z - n), & P_w &= -b(q)n - \mu, & P_q &= \Delta b n z, \\ G_n &= -\Delta r, & G_w &= -\Delta r - \Delta a, & G_q &= 0. \end{aligned}$$

The full reaction Jacobian in variables (n, w, q) is therefore

$$J = \begin{pmatrix} R - nr(q) & n[-r(q) - a(q)] & nG \\ w b(q)(z - n) & P + w[-b(q)n - \mu] & w \Delta b n z \\ -\nu q(1 - q)\Delta r & -\nu q(1 - q)(\Delta r + \Delta a) & \nu(1 - 2q)G \end{pmatrix}.$$

At an interior compensation-branch equilibrium, $R = P = G = 0$, giving the simplified branch Jacobian

$$J_* = \begin{pmatrix} -n^* r(q^*) & -n^*[r(q^*) + a(q^*)] & 0 \\ w^* b(q^*)(z^* - n^*) & -w^*[b(q^*)n^* + \mu] & w^* \Delta b n^* z^* \\ -\nu q^*(1 - q^*)\Delta r & -\nu q^*(1 - q^*)(\Delta r + \Delta a) & 0 \end{pmatrix}.$$

For $\det(\lambda I - J) = \lambda^3 + A_1 \lambda^2 + A_2 \lambda + A_3$, the coefficients were computed as

$$A_1 = -\text{tr}(J), \quad A_2 = \frac{1}{2} [(\text{tr} J)^2 - \text{tr}(J^2)], \quad A_3 = -\det(J).$$

Table 1 reports the Routh–Hurwitz terms and the minimum margin at the target stresses.

Table 1: Routh–Hurwitz margins for the linear compensation branch at target stresses. Values are copied to `results/roy_ode_compensation_routh_hurwitz_margins.csv`.

	s	A_1	A_2	A_3	$A_1 A_2 - A_3$	min margin
	0	3.9468	0.33673	0.003540	1.3255	0.003540
	0.069448242	4.3241	0.27528	0.004006	1.1863	0.004006
	0.11765625	4.5861	0.21486	0.003566	0.98178	0.003566
	0.1584375	4.8077	0.15238	0.002706	0.72989	0.002706
	0.16486816	4.8426	0.14158	0.002529	0.68308	0.002529
	0.175	4.8976	0.12403	0.002228	0.60525	0.002228

The dense branch-stability scan in `results/roy_ode_compensation_dense_stability_scan.csv` evaluates 1001 stress values across the open interior branch interval, excluding the trait-boundary endpoints by 10^{-8} . All sampled points satisfy the Routh–Hurwitz inequalities and eigenvalue stability; the minimum margin is small because it approaches zero near the trait-boundary endpoints.

S3. Additional ODE basin maps

The main text focuses on the derived homogeneous compensation branch and its stability. Additional ODE diagnostics support the basin interpretation and mechanism. Figure 3 shows representative ODE time series and selection/growth diagnostics. Figure 4 shows the ODE q_0 – w_0 basin map used to relate initial defense and predator density to persistent, extinct, and transient outcomes. Figure 5 shows selection-growth phase diagnostics and numerical equilibrium stability.

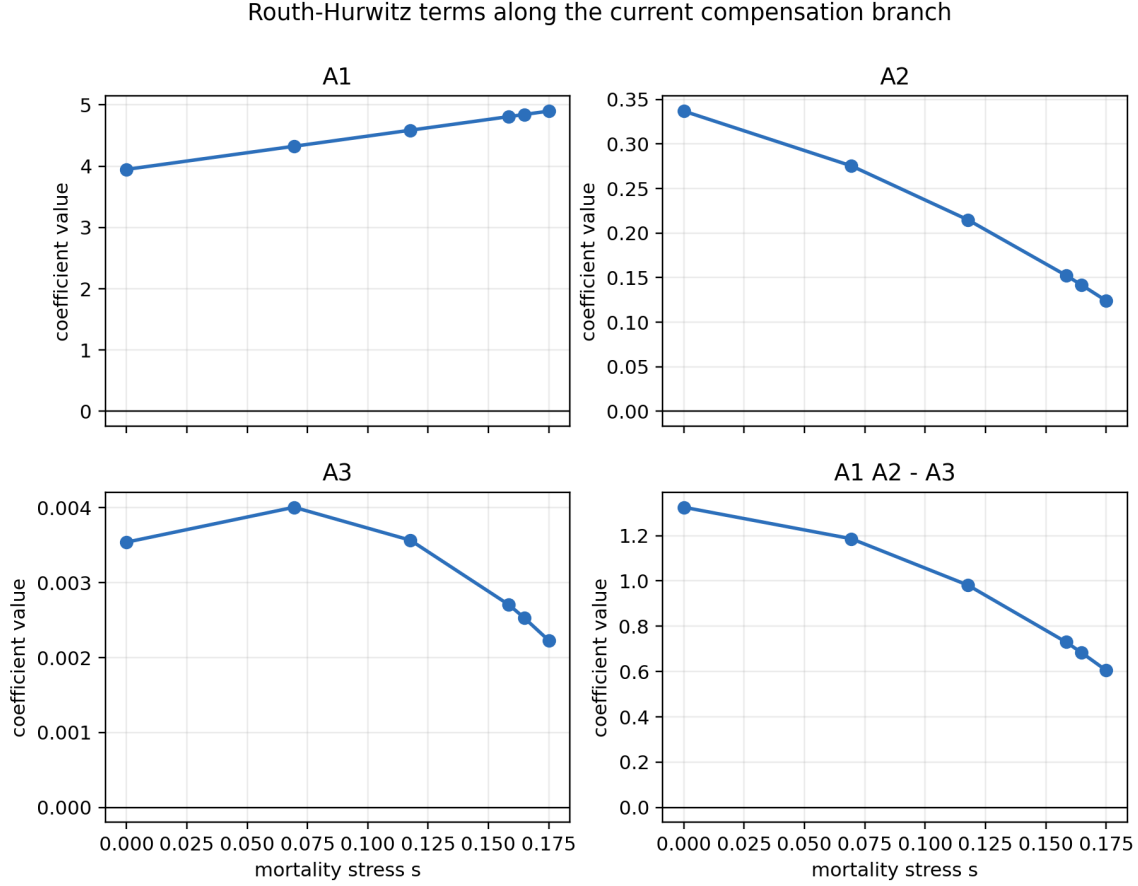


Figure 1: Full Routh–Hurwitz coefficient diagnostics for the linear compensation branch. The main text shows the minimum margin and maximum eigenvalue real part; this supporting figure keeps the individual A_1 , A_2 , A_3 , and $A_1 A_2 - A_3$ terms visible for reproducibility.

Table 2: Dense local-stability scan summary for the open linear compensation branch.

Metric	Value
Dense grid points	1001
Stress interval sampled	$(-0.113138879, 0.232444434)$
All dense points RH/eigenvalue stable	True
Minimum dense RH margin	4.65×10^{-10}
Maximum dense eigenvalue real part	-1.25×10^{-9}

Dense local-stability scan across the open compensation branch

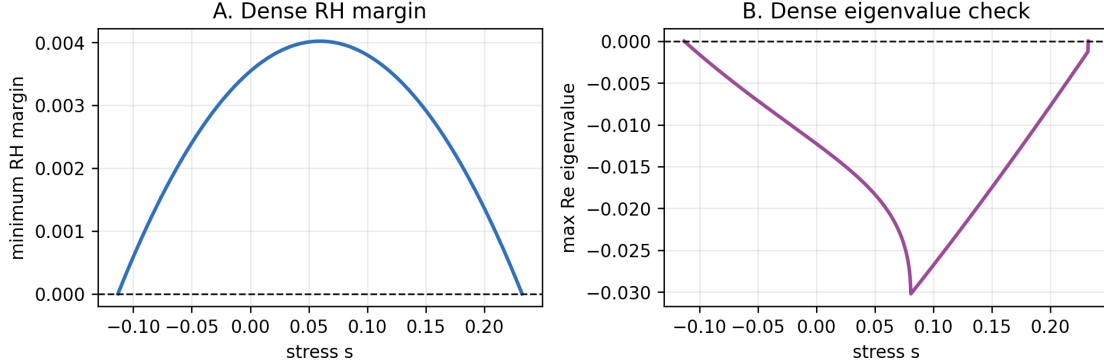


Figure 2: Dense local-stability scan across the open compensation branch. Margins approach zero near the trait-boundary endpoints, so this is evidence for sampled open-interval local stability, not endpoint stability.

S4. PDE non-homogeneous perturbation diagnostics

The main manuscript reports the conclusion from spatial-mode stability and long-horizon basin-resolution analysis. The supporting PDE diagnostics are shown here. Figure 6 summarizes additional spatial-stability views. Figure 7 shows the sampled continuous- λ scan over the mode-equivalent range $i, j = 0, \dots, 64$, sampled at 600 equally spaced λ values. Figure 8 adds a denser semi-analytic modal Routh–Hurwitz diagnostic over the same λ -range. This diagnostic computes the cubic coefficients of $J_F(U^*) - \lambda D$ over a sampled grid; it is stronger than reporting eigenvalues alone but remains sampled evidence rather than a proof for every λ . Figures 9 and 10 show representative non-homogeneous initial and final fields. Figures 11 and 12 show mean dynamics and spatial coefficient-of-variation diagnostics for the targeted non-homogeneous tests. Figures 13–15 show the long-horizon diagnostics for the finite-horizon basin-changing cases.

Because q is a frequency variable, the numerical implementation enforces admissibility by clipping q to $[0, 1]$ after explicit PDE updates. This clipping is a numerical admissibility constraint, not a proof that the phenomenological PDE closure preserves $q \in [0, 1]$ without enforcement. Finite-amplitude PDE conclusions should therefore be interpreted under this implementation. The bounds audit in `results/roy_pde_q_bounds_audit.csv` checks saved post-update field snapshots rather than raw pre-clipping trial values. All audited saved fields have $q \in [0, 1]$ within tolerance. Table 3 summarizes the audit by diagnostic group.

Table 3: q -bounds audit for saved PDE diagnostic field files.

Diagnostic group	Files	min q	max q
Representative PDE fields	3	8.17×10^{-7}	0.9000
Non-homogeneous PDE fields	5	0	0.3142
Long-horizon non-homogeneous fields	2	2.78×10^{-12}	0.2142
Nonlinear trade-off PDE fields	12	0	0.6615

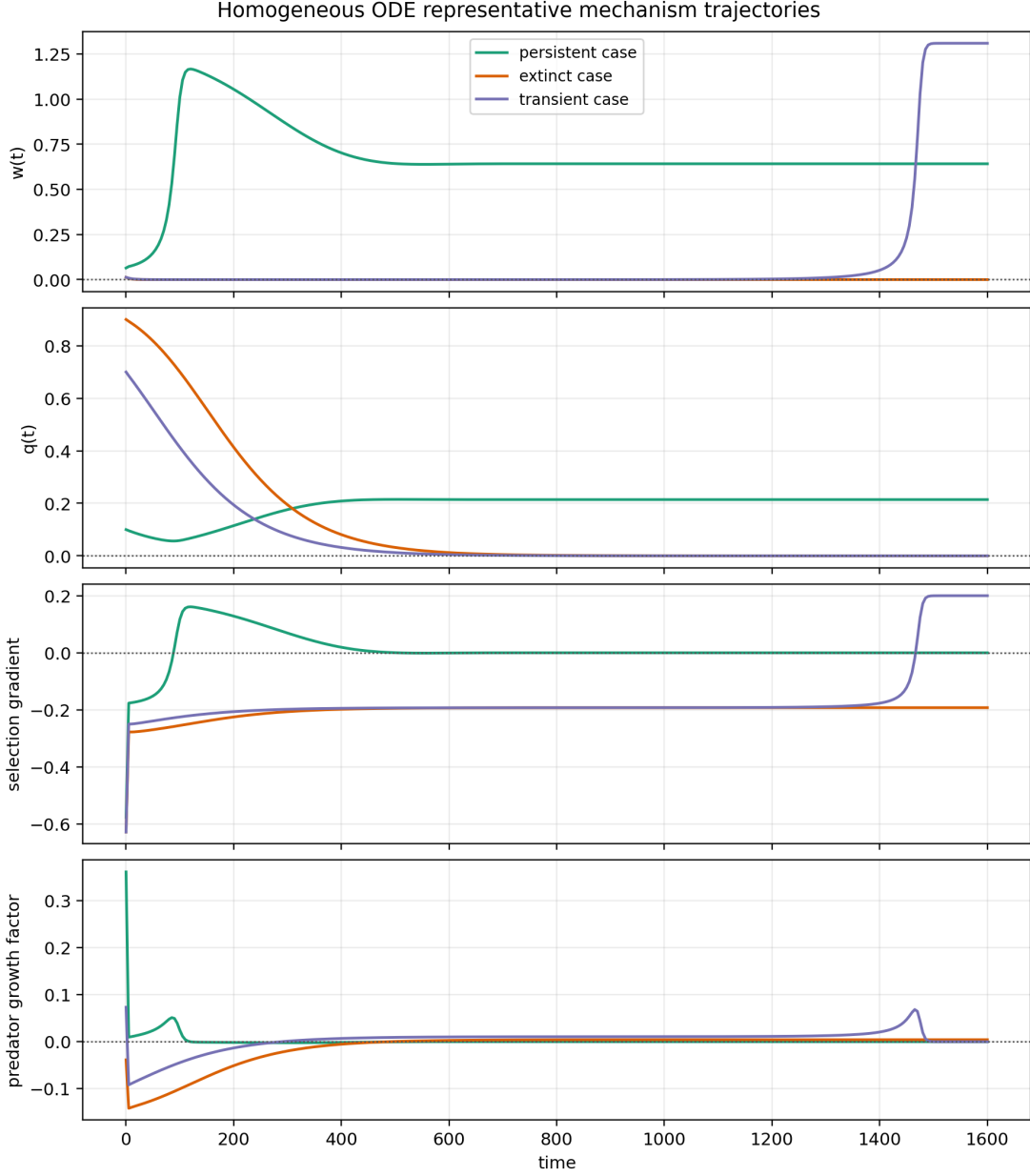


Figure 3: Representative ODE trajectories for predator density, defense frequency, selection-gradient diagnostics, and predator-growth diagnostics. These panels support the homogeneous mechanism interpretation but do not replace the analytic branch derivation.

Table 4: Dense sampled modal Routh–Hurwitz summary for $J_F(U^*) - \lambda D$. The λ -grid uses 2001 equally spaced values over the same mode-equivalent range as the main continuous- λ scan.

Stress s	Min modal RH margin	Max modal growth	Stable samples	RH/eigen disagreements
0.117656	3.57×10^{-3}	-2.36×10^{-2}	2001/2001	0
0.158438	2.71×10^{-3}	-1.59×10^{-2}	2001/2001	0
0.164868	2.53×10^{-3}	-1.47×10^{-2}	2001/2001	0
0.175000	2.23×10^{-3}	-1.27×10^{-2}	2001/2001	0

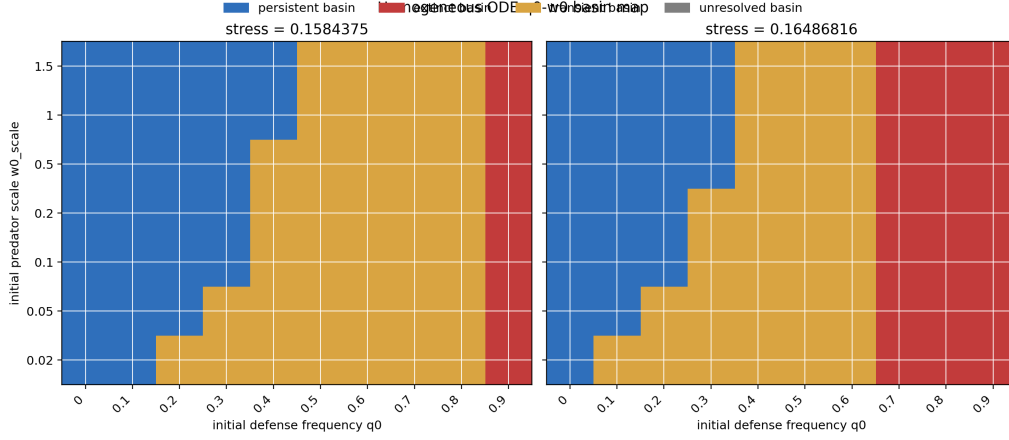


Figure 4: ODE basin map on the targeted q_0 - w_0 grid. This figure supports basin dependence in the homogeneous reaction system and is not an exhaustive basin-boundary refinement.

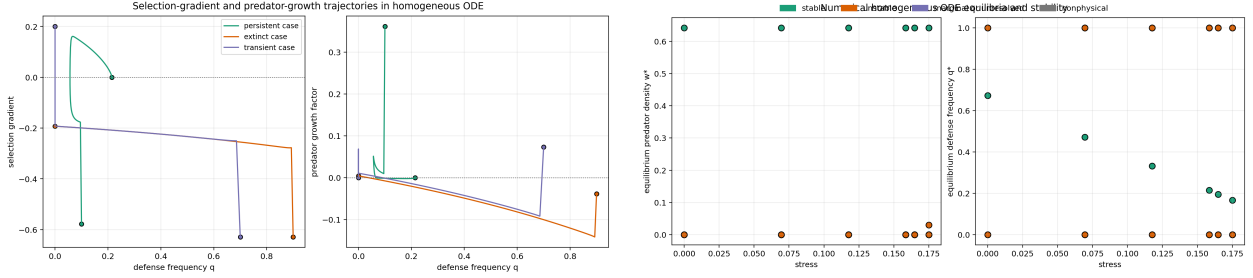


Figure 5: Additional homogeneous mechanism diagnostics. Left: selection-gradient and predator-growth relationships for representative cases. Right: numerical equilibria and stability labels across tested stresses.

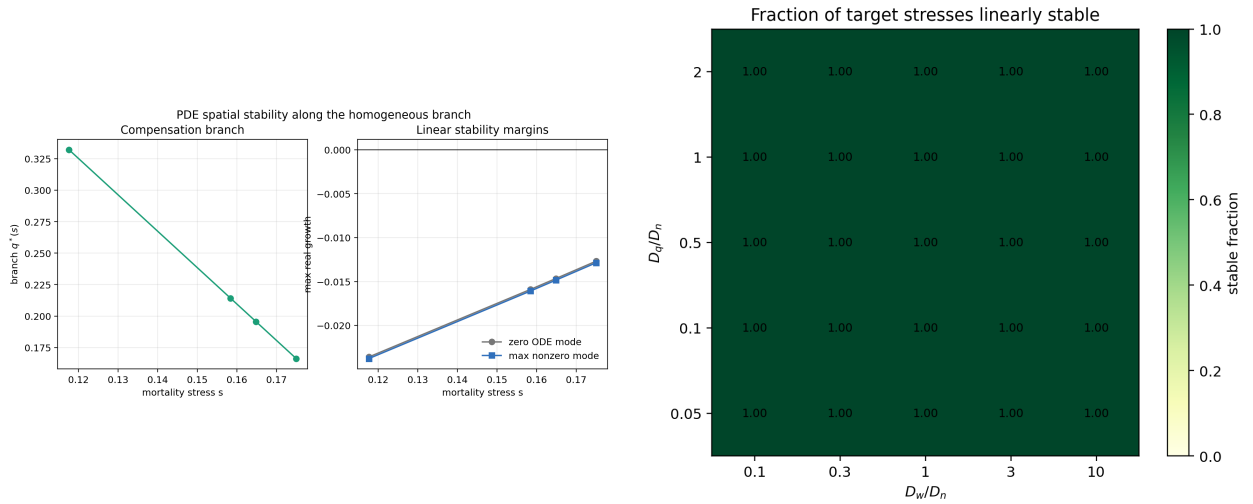


Figure 6: Additional linear PDE spatial-stability diagnostics for the tested branch states and diffusion-ratio grid.

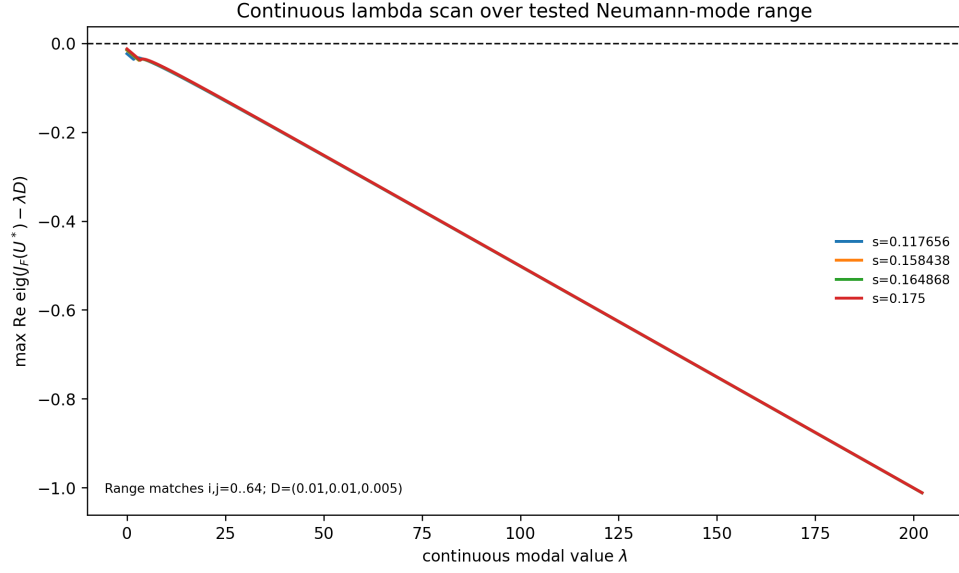


Figure 7: Sampled continuous- λ scan for $J_F(U^*) - \lambda D$ over 600 equally spaced λ values in the mode-equivalent range corresponding to $i, j = 0, \dots, 64$.

Dense sampled modal stability margins: min RH margin=0.00223, max growth=-0.0127

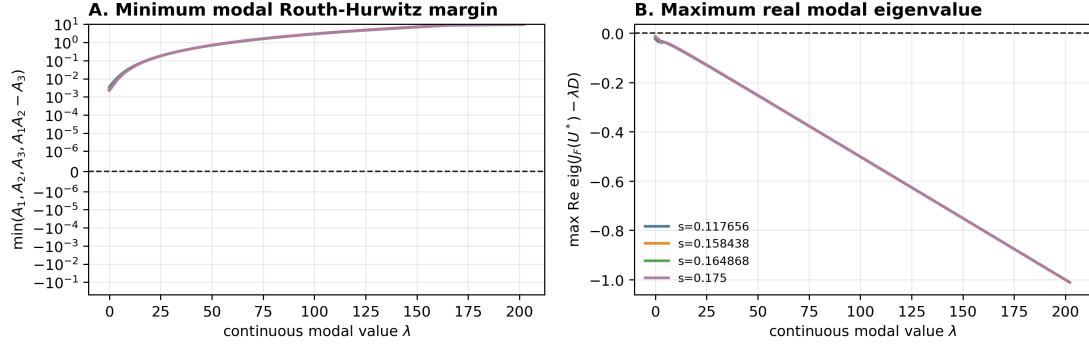


Figure 8: Dense sampled modal Routh–Hurwitz margins for $J_F(U^*) - \lambda D$. The calculation samples 2001 λ values over the mode-equivalent range corresponding to $i, j = 0, \dots, 64$. Positive margins and negative maximum real eigenvalues support sampled modal stability over this range.

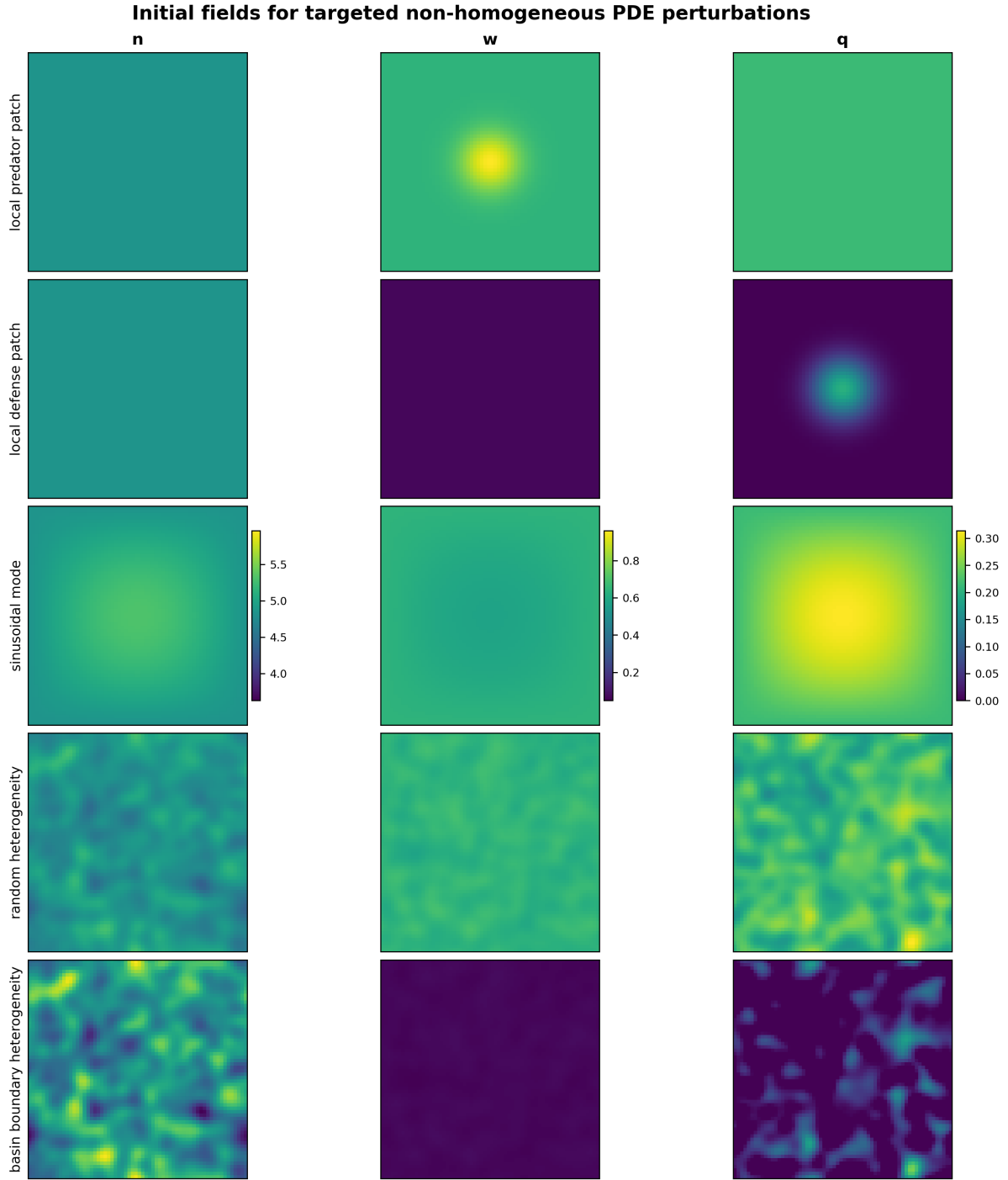


Figure 9: Representative non-homogeneous initial PDE fields for targeted finite-amplitude perturbations. Color scales are variable-specific and shared across rows within this figure; initial and final field figures are scaled independently.

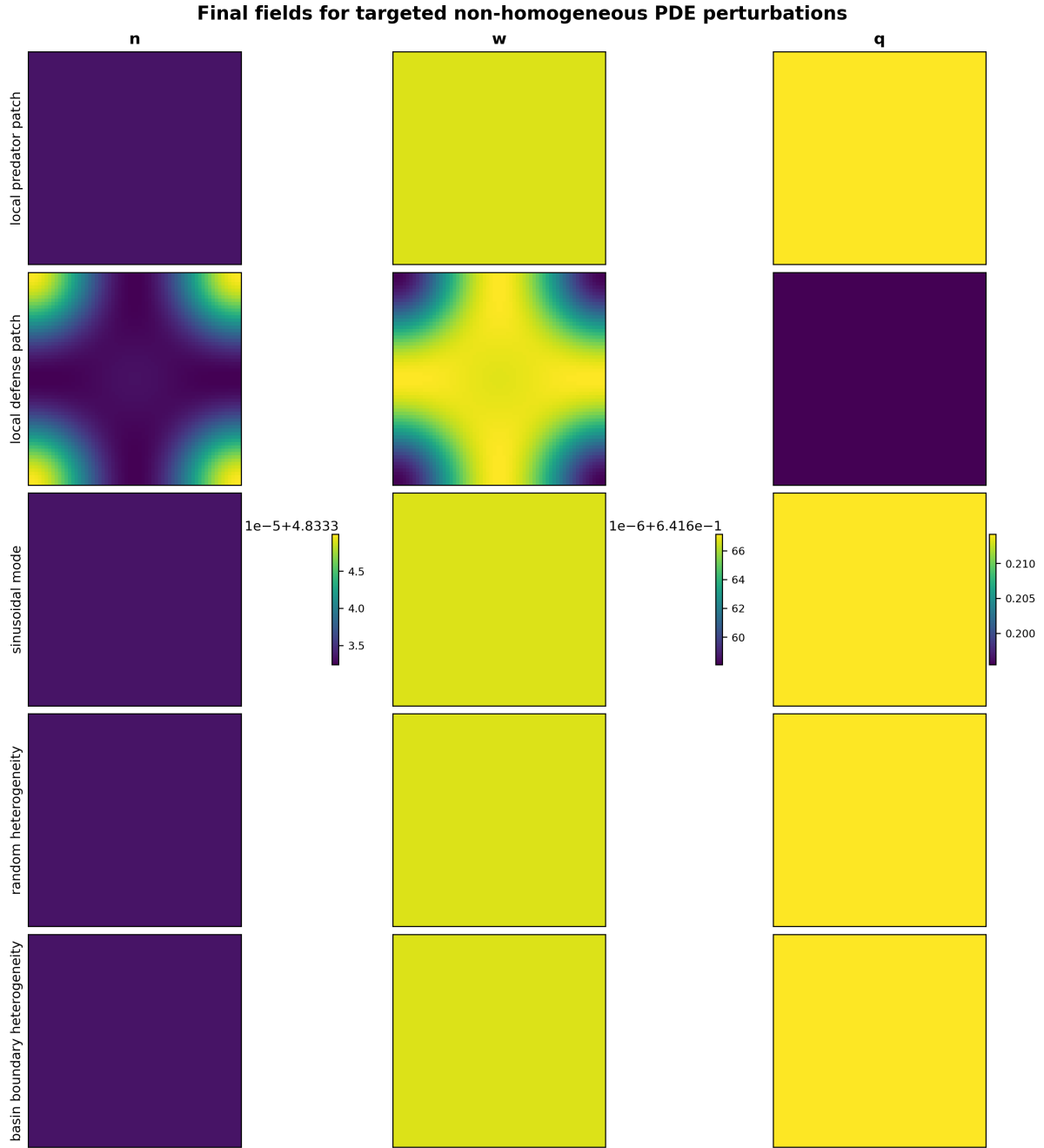


Figure 10: Final PDE fields for representative targeted non-homogeneous perturbation tests. Color scales are variable-specific and shared across rows within this figure; initial and final field figures are scaled independently.

Targeted non-homogeneous PDE mean dynamics

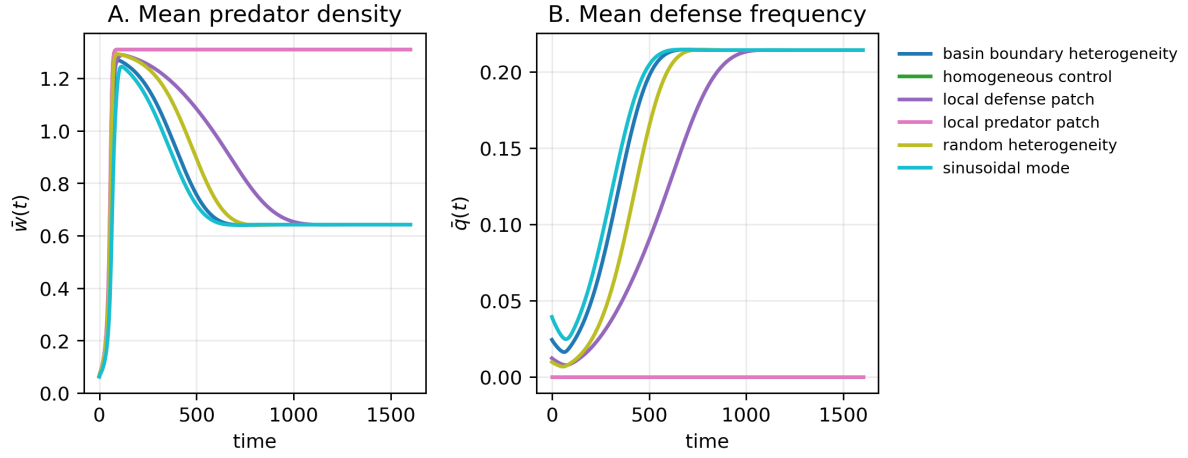


Figure 11: Mean dynamics for targeted finite-amplitude non-homogeneous PDE tests and matched homogeneous controls.

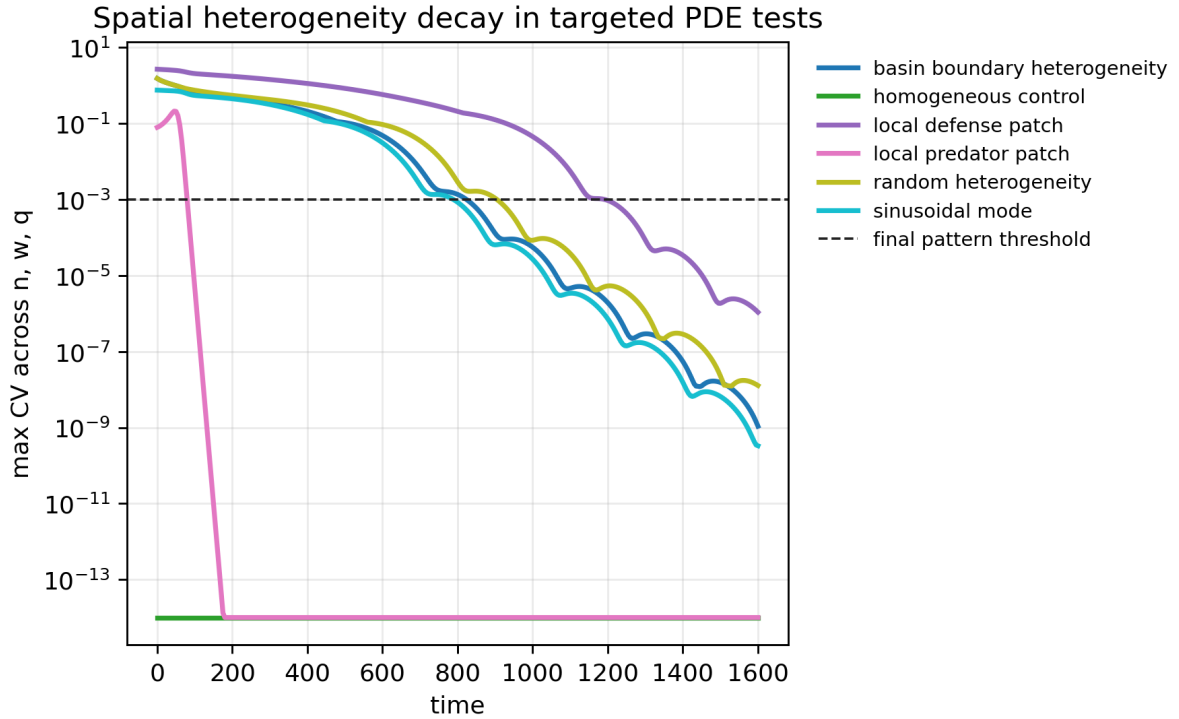


Figure 12: Spatial coefficient-of-variation diagnostics for targeted finite-amplitude non-homogeneous PDE tests.

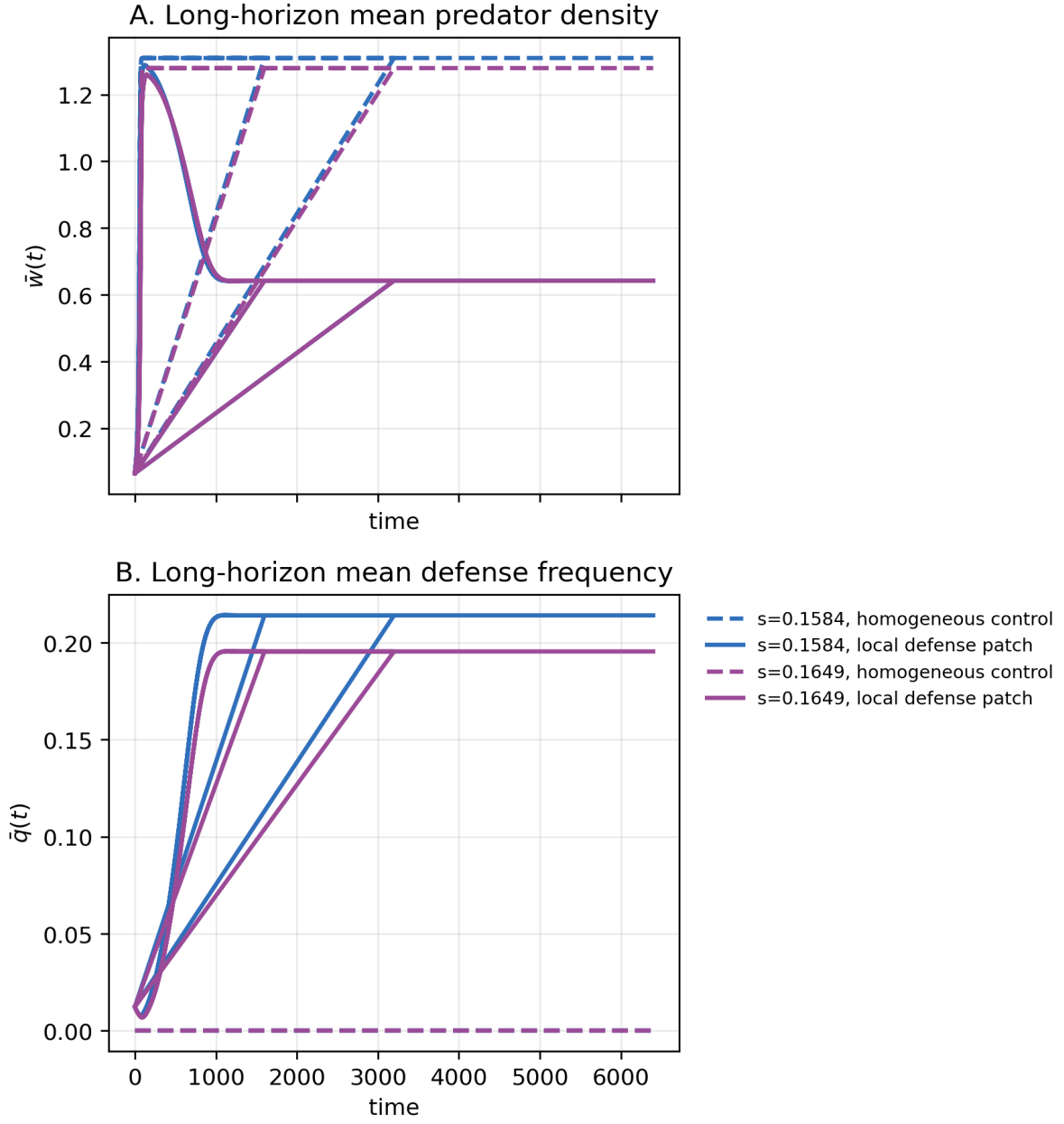


Figure 13: Long-horizon mean time series for the finite-horizon basin-changing non-homogeneous cases. The cases are the targeted cases selected from the earlier finite-amplitude tests, not a broad PDE scan.

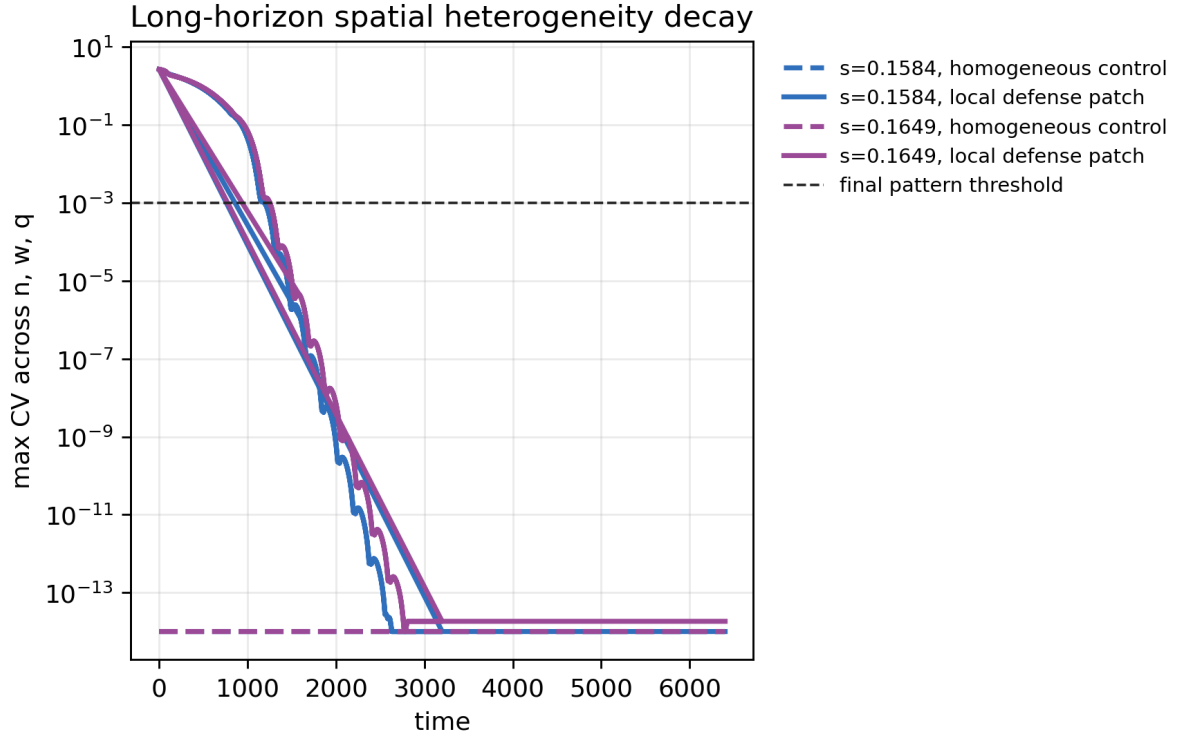


Figure 14: Long-horizon spatial heterogeneity metrics for the followed basin-changing cases. These diagnostics support spatial-pattern decay in the followed cases.

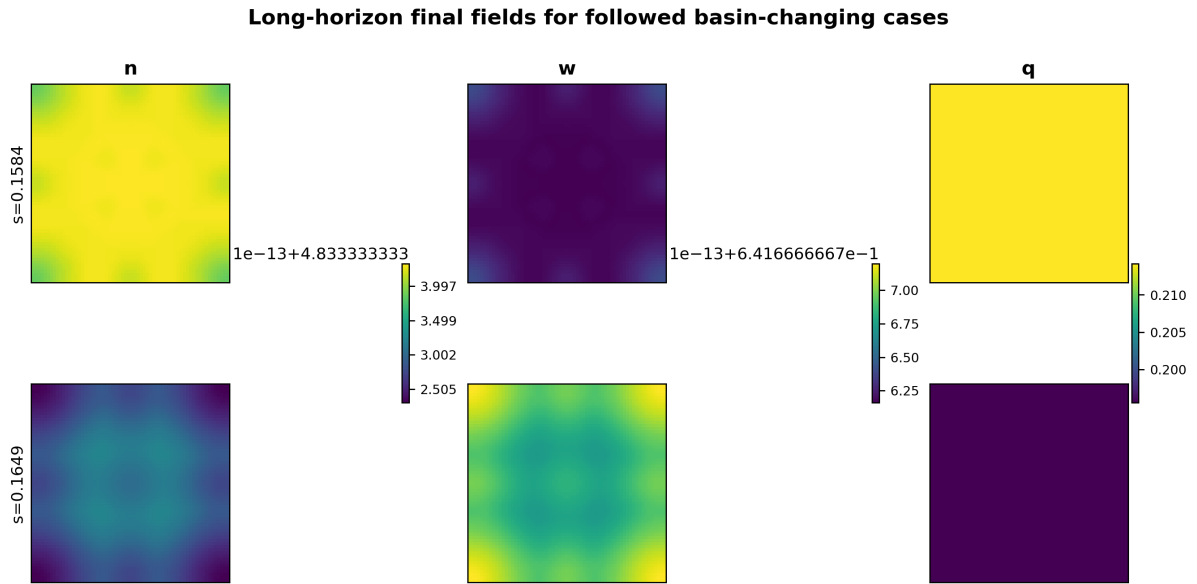


Figure 15: Long-horizon final fields for the basin-changing cases selected for longer-horizon analysis. These diagnostics support resolution without persistent spatial patterning for the followed cases.

S5. Nonlinear trade-off shape-grid details

The nonlinear extension uses endpoint-preserving shape functions $f_\gamma(q) = q^\gamma$ with $\gamma \in \{0.5, 1, 2\}$. The purpose is to test whether the compensation mechanism survives selected concave, convex, and mixed trade-off shapes, not to scan all possible nonlinear functions. Figure 16 shows selected nonlinear branch curves and shape-grid summaries. Figures 17–19 show selected nonlinear ODE basin maps, PDE spatial-stability diagnostics, and non-homogeneous PDE diagnostics.

The no-compensation and unresolved shape classes identify trade-off geometries where the existing diagnostics did not support the compensation mechanism. Table 5 lists these cases from `results/roy_nonlinear_tradeoff_failure_modes.csv`. Four shapes had no feasible branch at the four target stresses. One shape had branches at three target stresses but no target stress was classified as stable in the existing diagnostics. These failure modes show dependence on trade-off geometry; they are not expanded into a broader nonlinear scan here.

The nonlinear mechanism can fail at several distinct steps. First, the selection-gradient geometry is controlled by $c(q) = r'(q)/a'(q)$; if this ratio is not positive or gives nonpositive $z(q)$, $w(q)$, or $n(q)$, an interior compensation state is infeasible. Second, even when candidate states are feasible, the stress map $s(q) = b(q)n(q)z(q) - m - \mu w(q)$ must overlap the rescue-relevant stresses being tested. Third, conversion curvature in $b(q)$ controls whether a lower-defense shift produces enough predator-growth compensation as stress increases. Fourth, branch existence alone is not sufficient because the candidate branch must also be locally stable and reachable from the initial conditions used in the basin diagnostics. The controlled shape grid therefore separates absence of compensation into feasibility, stress-overlap, conversion-response, stability, and reachability limitations rather than treating all failures as the same mechanism; this is an interpretation of the tested grid, not a general theorem.

Table 5: No-compensation and unresolved nonlinear trade-off shapes from the existing shape-grid diagnostics.

γ_r	γ_a	γ_b	Class	Reason from existing diagnostics
0.5	1.0	0.5	no compensation	No feasible branch at target stresses
0.5	2.0	0.5	no compensation	No feasible branch at target stresses
1.0	2.0	0.5	no compensation	No feasible branch at target stresses
1.0	2.0	1.0	no compensation	No feasible branch at target stresses
2.0	1.0	0.5	unresolved	Branch found at three target stresses, but no stable target stress

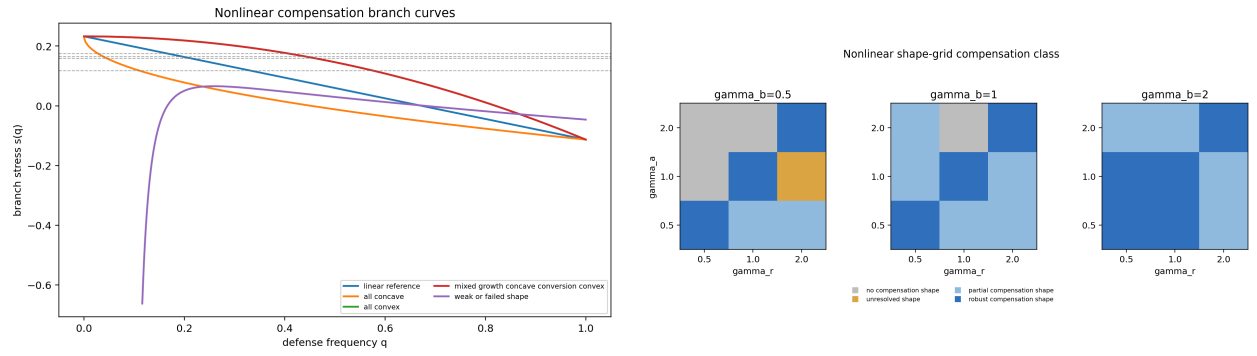


Figure 16: Nonlinear branch curves and shape-grid summary. These figures support selected non-linear compensation branches within the controlled grid.

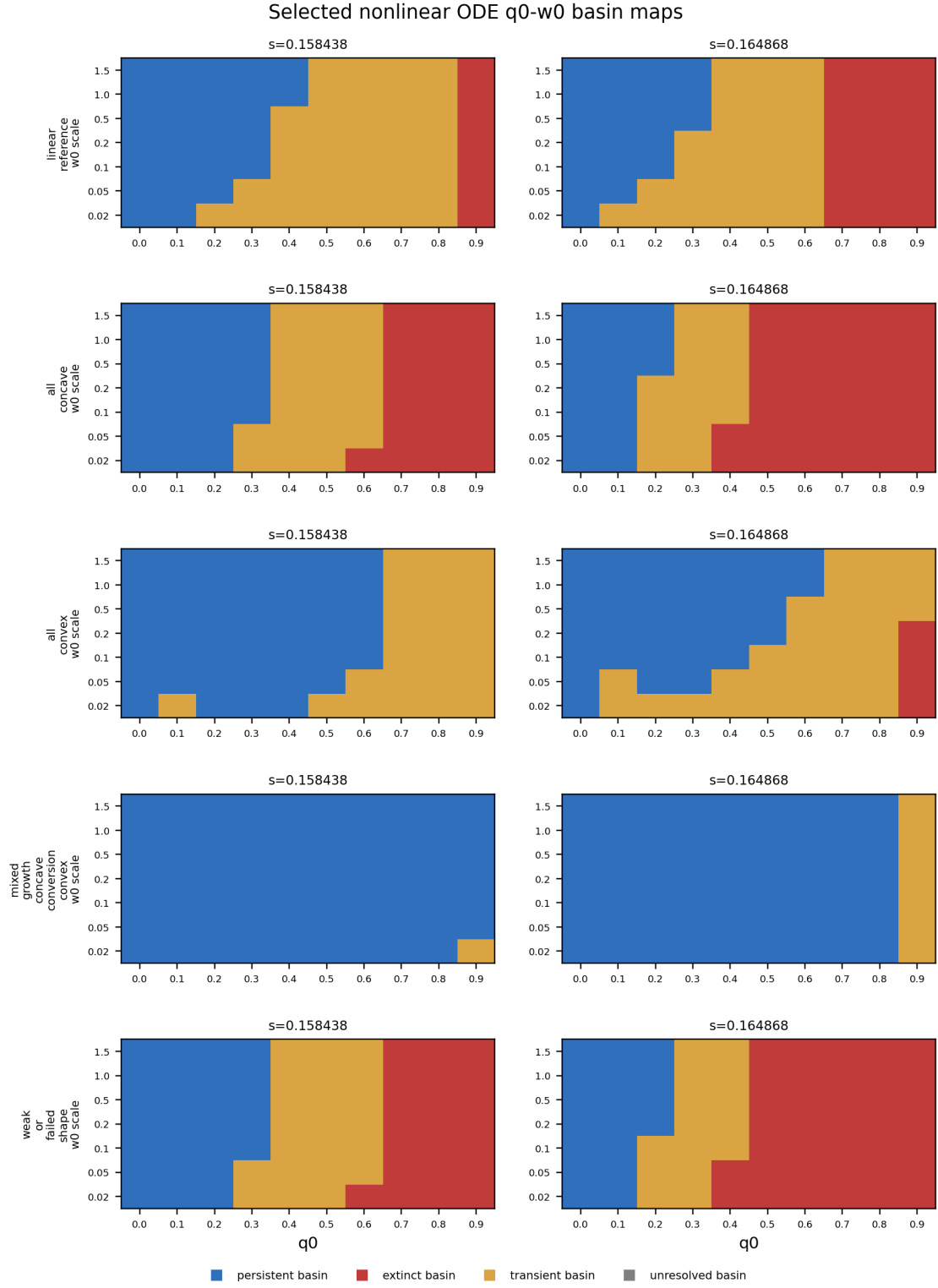


Figure 17: Selected nonlinear ODE basin maps for the controlled shape-grid extension.

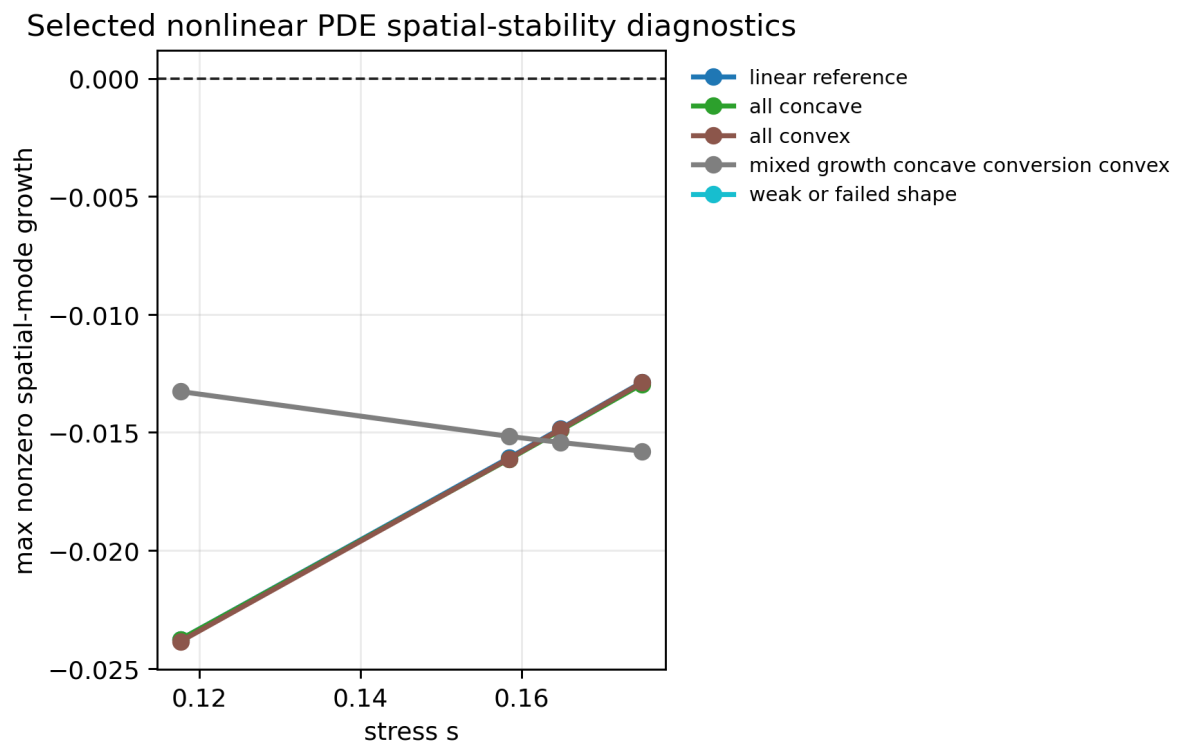


Figure 18: Selected nonlinear PDE spatial-stability diagnostics for selected nonlinear branches.

Selected nonlinear non-homogeneous PDE diagnostics

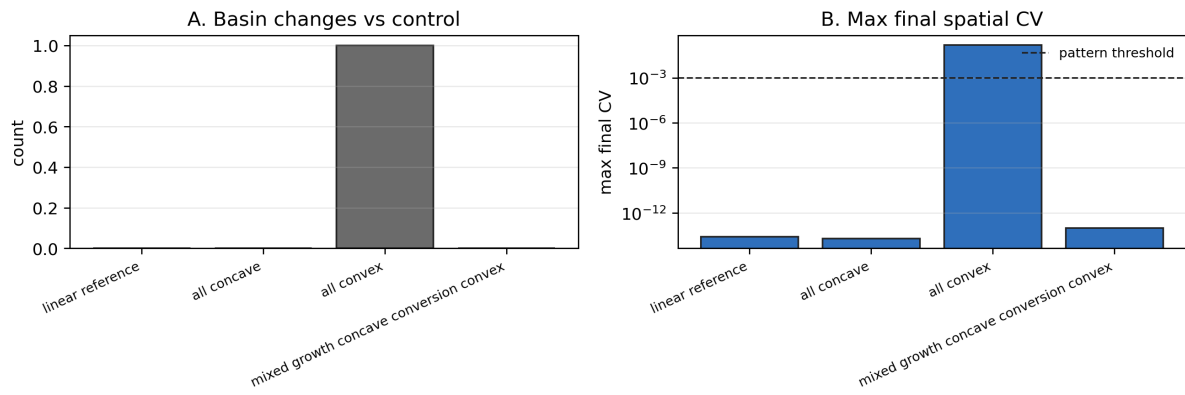


Figure 19: Selected nonlinear non-homogeneous PDE diagnostics for targeted finite-amplitude checks.

S6. Ecological-review sensitivity diagnostics

Figure 20 reports a finite-time sensitivity diagnostic for the defense-evolution rate ν . These runs use the same ODE equations and baseline parameterization except for ν , start from the pre-stress branch initial state, and test stresses $s \in \{0.08, 0.10, 0.1191, 0.14, 0.16\}$. The diagnostic is not a new rescue-window claim. It shows that finite-time rescue can depend on evolutionary rate even though the branch equilibrium itself is independent of ν . Because the baseline classifier is tail-based, some deterministic trajectories can be classified as persistent after dropping below the extinction threshold. For ecological interpretation, the sensitivity diagnostic therefore also reports whether $w(t)$ crossed the extinction threshold. At the representative rescue stress $s = 0.1191$, $\nu = 0.005$ loses predators before adaptation completes, $\nu = 0.01$ recovers mathematically after crossing below the threshold, and $\nu \geq 0.025$ remains predator-persistent without threshold crossing under the tested horizon.

Finite-time evolutionary-rate sensitivity from the pre-stress branch state

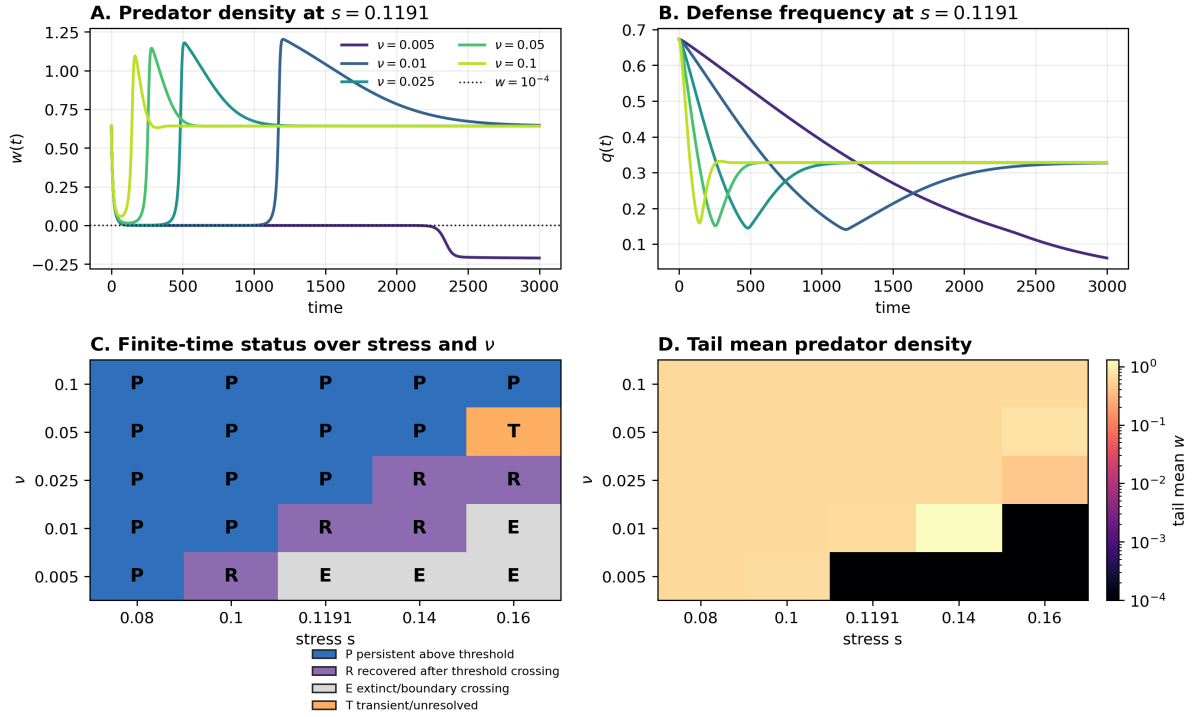


Figure 20: Finite-time evolutionary-rate sensitivity diagnostic. Panels A and B show $w(t)$ and $q(t)$ at $s = 0.1191$; the dotted line marks the extinction threshold. Panel C reports ecological finite-time status: P, persistent without threshold crossing; R, tail-persistent recovery after threshold crossing; E, extinct or boundary crossing; T, transient or unresolved. Panel D shows tail mean predator density across the small stress- ν grid.

Figure 21 reports a linear modal sensitivity diagnostic for the phenomenological q -diffusion coefficient. The calculation varies $D_q \in \{0, 0.001, 0.005, 0.01, 0.05\}$, keeps $D_n = D_w = 0.01$, and evaluates the same sampled continuous- λ range used in the main spatial-stability evidence. This is a linear diagnostic only; it does not add PDE simulations or establish finite-amplitude domain robustness.

Linear modal sensitivity to phenomenological q -diffusion

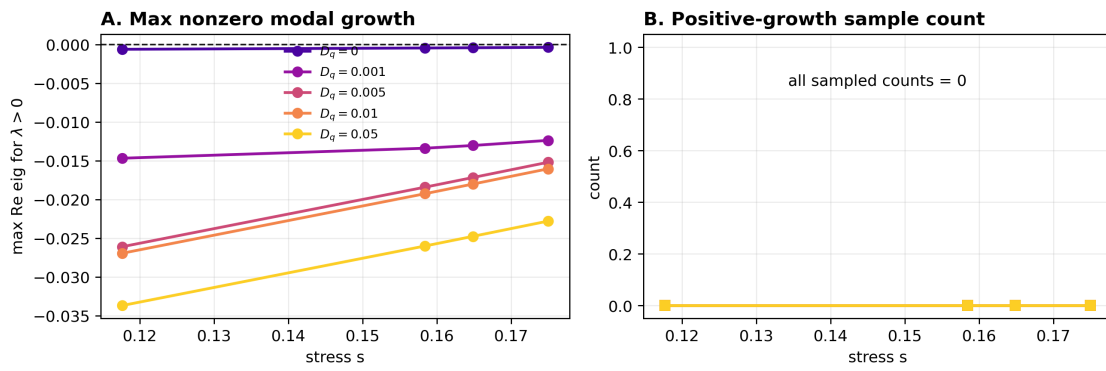


Figure 21: Linear modal sensitivity to q -diffusion. Panel A shows the maximum sampled growth rate for positive λ . Panel B reports the number of sampled positive-growth values.

S7. Additional figure list and CSV source mapping

Table 6 maps the main analysis scripts to the CSV outputs used by the manuscript package. Table 7 maps the main claims to source CSVs and representative figures. The status column records whether each item is a main result or supporting diagnostic evidence. All script names in these tables are relative to `experiments/`; all CSV names are relative to `results/`. For readability, the tables use numbered script short names and compact CSV names; the full paths are present in the repository.

Table 6: Script-to-CSV mapping for reproducibility.

Script	CSV outputs used here
Step 23: compensation conditions	<code>compensation_conditions_summary</code> ; <code>compensation_condition_grid</code>
Step 24: Routh–Hurwitz	<code>routh_hurwitz_current</code> ; <code>routh_hurwitz_summary</code>
Step 25: PDE spatial and heterogeneous tests	<code>pde_spatial_modes_current</code> ; <code>pde_spatial_nonlinear_decision</code>
Step 26: long-horizon basin-resolution analysis	<code>pde_long_horizon_decision</code> ; summary and time-series CSVs
Step 27: nonlinear trade-offs	<code>nonlinear_compensation_decision</code> ; <code>nonlinear_shape_summary</code> ; <code>nonlinear_branch_recovery</code>
Step 28: counterfactual and modal scan	<code>no_evolution_counterfactual_summary</code> ; stress-sweep and time-series CSVs; <code>pde_lambda_scan_summary</code> ; <code>rh_margins</code>
Step 29: manuscript audit	<code>compensation_branch_value_audit</code> ; <code>compensation_branch_reference_doi_audit</code> ; <code>nonlinear_tradeoff_failure_modes</code>
Step 30: ecological-review sensitivities	<code>ode_nu_sensitivity_summary</code> ; <code>pde_dq_modal_sensitivity_summary</code>
Step 31: mathematical and classification tightening	<code>ode_compensation_dense_stability_scan</code> ; <code>pde_q_bounds_audit</code>
Step 32: spatial and nonlinear robustness clarification	<code>pde_compensation_modal_routh_hurwitz_scan</code> ; <code>pde_compensation_modal_routh_hurwitz_summary</code>

Table 7: Claim-to-source mapping for the manuscript package.

Claim	Source CSV	Representative figure	Status
Frozen- q counterfactual verifies indirect rescue over a finite stress window	<code>no_evolution_counterfactual_summary</code> ; stress-sweep CSV	Main no-evolution figure	Main claim
Threshold-safe persistence gives a narrower ecological rescue window than the deterministic tail-based classifier	<code>no_evolution_counterfactual_summary</code> ; stress-sweep CSV	Main no-evolution figure	Main claim qualifier
Linear compensation branch conditions are satisfied in the current parameterization	<code>compensation_conditions_summary</code> ; <code>compensation_branch_value_audit</code>	Main branch figure; Fig. 5	Main claim
Routh–Hurwitz stability holds on the dense sampled open branch interval	<code>routh_hurwitz_summary</code> ; <code>ode_compensation_dense_stability_scan</code>	Main Routh–Hurwitz figure; Fig. 1; Fig. 2	Main claim
Spatial modes are stable for the current PDE coefficients over the tested range	<code>pde_spatial_nonlinear_decision</code> ; <code>pde_lambda_scan_summary</code>	Main spatial-stability figure; Fig. 6; Fig. 7	Main claim
Modal Routh–Hurwitz margins are positive over the dense sampled λ -grid for the target branch stresses	<code>pde_compensation_modal_routh_hurwitz_summary</code> ; modal RH scan CSV	Fig. 8; Table 4	Supporting diagnostic
Saved PDE diagnostic fields keep q within the unit interval after the solver update/clipping step	<code>pde_q_bounds_audit</code>	Table 3	Supporting diagnostic

Claim	Source CSV	Representative figure	Status
Long-horizon non-homogeneous basin changes resolve to homogeneous controls	<code>pde_long_horizon_decision</code>	Main long-horizon decision panel; Fig. 13; Fig. 14; Fig. 15	Main claim
Finite-time rescue can depend on ν	<code>ode_nu_sensitivity_summary</code>	Fig. 20	Supporting diagnostic
Linear q -diffusion sensitivity shows no positive sampled modal growth in the tested set	<code>pde_dq_modal_sensitivity_summary</code>	Fig. 21	Supporting diagnostic
Linear branch is recovered by nonlinear branch formulation	<code>nonlinear_branch_recovery</code>	Fig. 16	Supporting
Nonlinear shape grid includes robust, partial, no-compensation, and unresolved cases	<code>nonlinear_shape_summary</code> ; <code>nonlinear_tradeoff_failure_modes</code> ; <code>compensation_branch_value_audit</code>	Main nonlinear heatmap; Table 5; Fig. 16	Main claim

The main manuscript contains the primary compensation branch and stress-interval figure. This boxed note avoids duplicating that main-text figure in the supplement while keeping the claim-to-source map readable.

Figure 22: Pointer to the main compensation branch figure.

S8. Reproducibility checklist and archival TODO

The manuscript package was built in a local Windows/PowerShell environment with Python and L^AT_EX tools available on the system path. The analyses are script-driven and write CSV outputs before figures and PDFs are rebuilt. The exact Python package versions should be captured from the final submission environment using `python -m pip freeze` or an equivalent lockfile before archival; this supplement records the command sequence but does not claim a frozen computational environment.

Command sequence for the main manuscript package. The following commands regenerate the current review-facing outputs used by this manuscript package:

```
python experiments/28_roy_counterfactual_and_spatial_evidence.py --profile focused
python experiments/30_roy_ecological_review_sensitivities.py --profile focused
python experiments/31_roy_mathematical_classification_tightening.py --profile focused
python experiments/32_roy_spatial_nonlinear_robustness_clarification.py --profile focused
python experiments/29_roy_manuscript_audit.py --profile focused
latexmk -pdf -outdir=manuscript -interaction=nonstopmode -halt-on-error manuscript/roy_compens
latexmk -pdf -outdir=manuscript -interaction=nonstopmode -halt-on-error manuscript/roy_compens
git diff --check
```

Archival TODO. Before journal submission, create an archival release on Zenodo or OSF containing the repository snapshot, generated CSV outputs, figure files, PDFs, and a machine-readable environment record. The archive should document the git commit hash, the script command sequence above, and the fact that the targeted PDE perturbation outputs are diagnostics rather than exhaustive spatial scans.